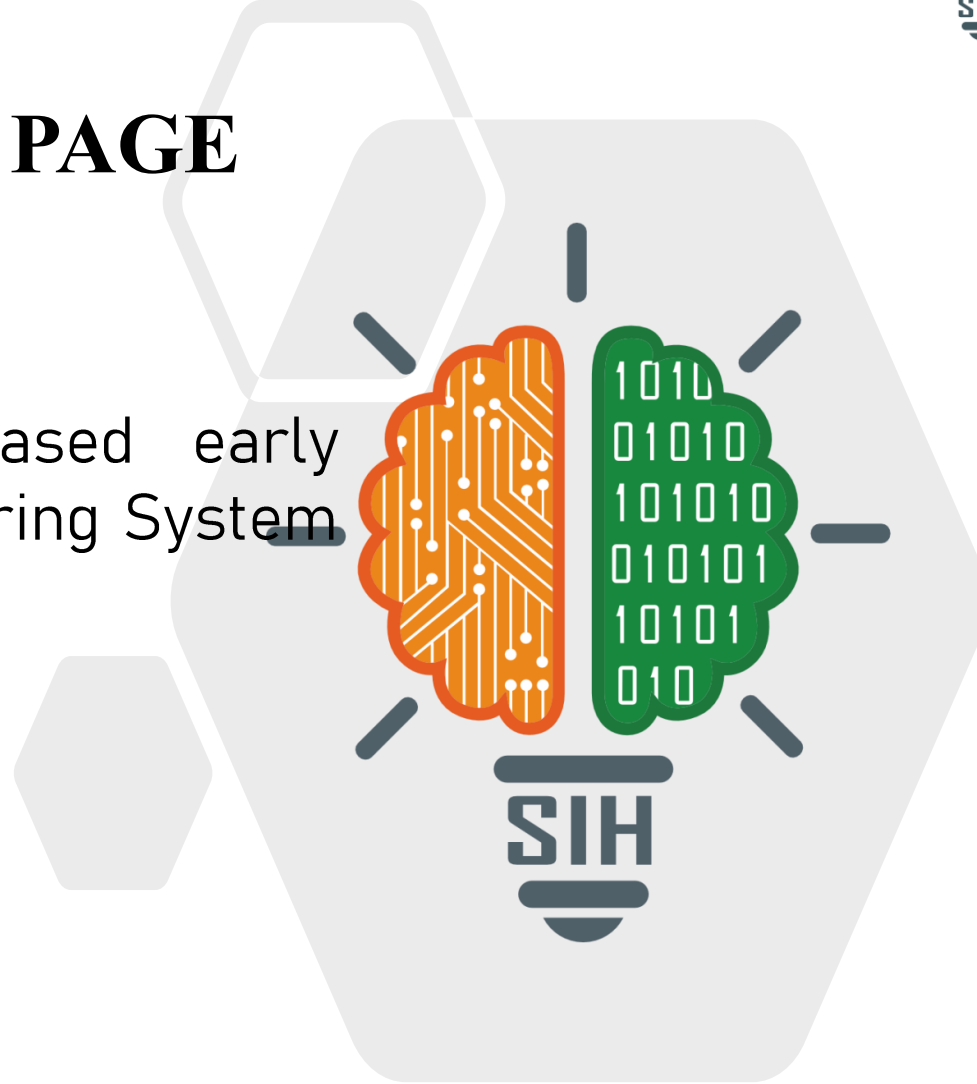
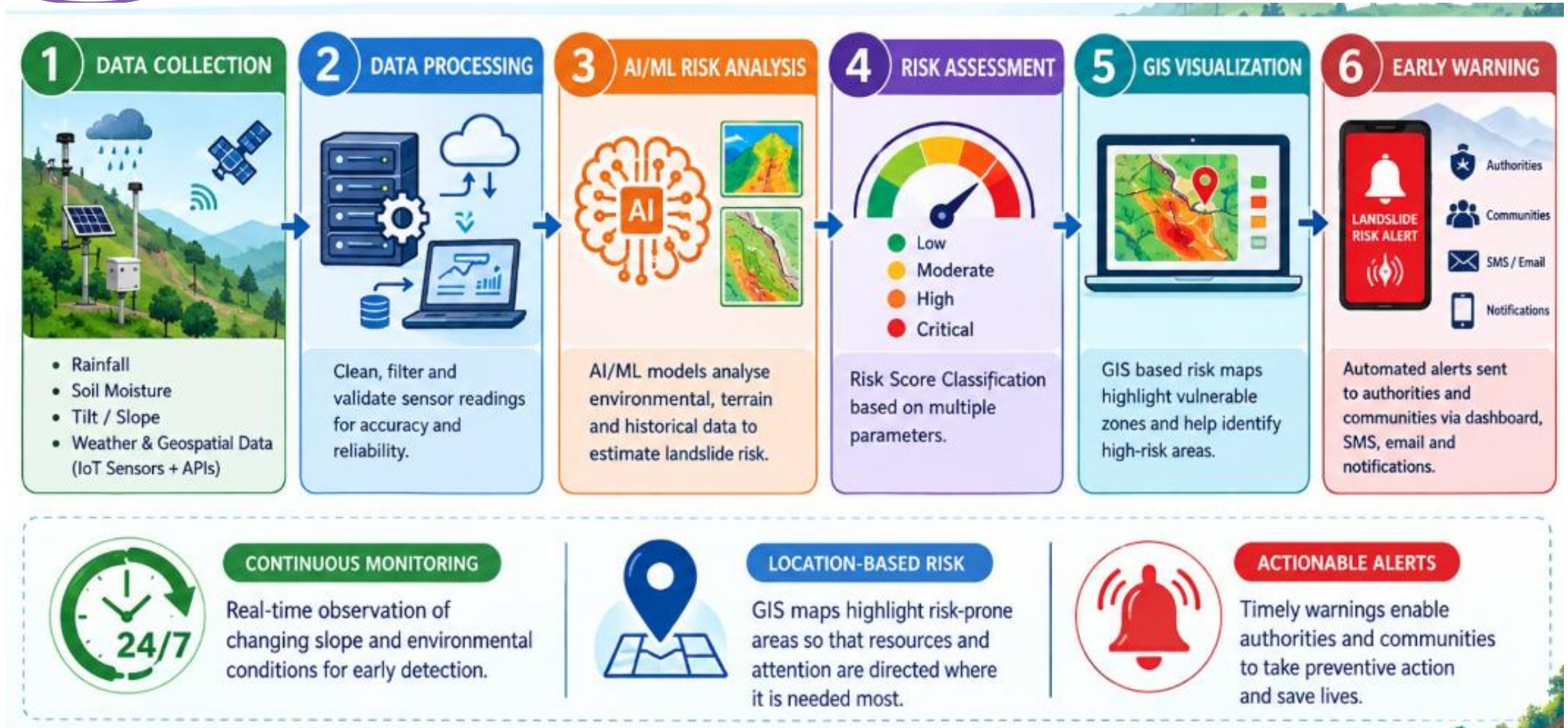


TITLE PAGE

- **Problem Statement ID** –SIH26001
- **Problem Statement Title-** AI-Based early warning and landslide Risk Monitoring System in NER
- **Theme-** Disaster Management
- **PS Category-** Software
- **Team ID-**
- **Team Name** –Geo-guardians





Technologies Used



Frontend: React.js
(HTML, CSS, JavaScript)



Backend: Node.js + Express.js



AI/ML: Python + Scikit-learn
(Prediction & Analysis)



Database: MongoDB
(Real-time data storage)



IoT: ESP32 + Rainfall, Soil
Moisture & Tilt Sensors



GIS: Interactive maps &
risk heatmaps



Communication: Wi-Fi / LoRa
(Data transfer & alerts)

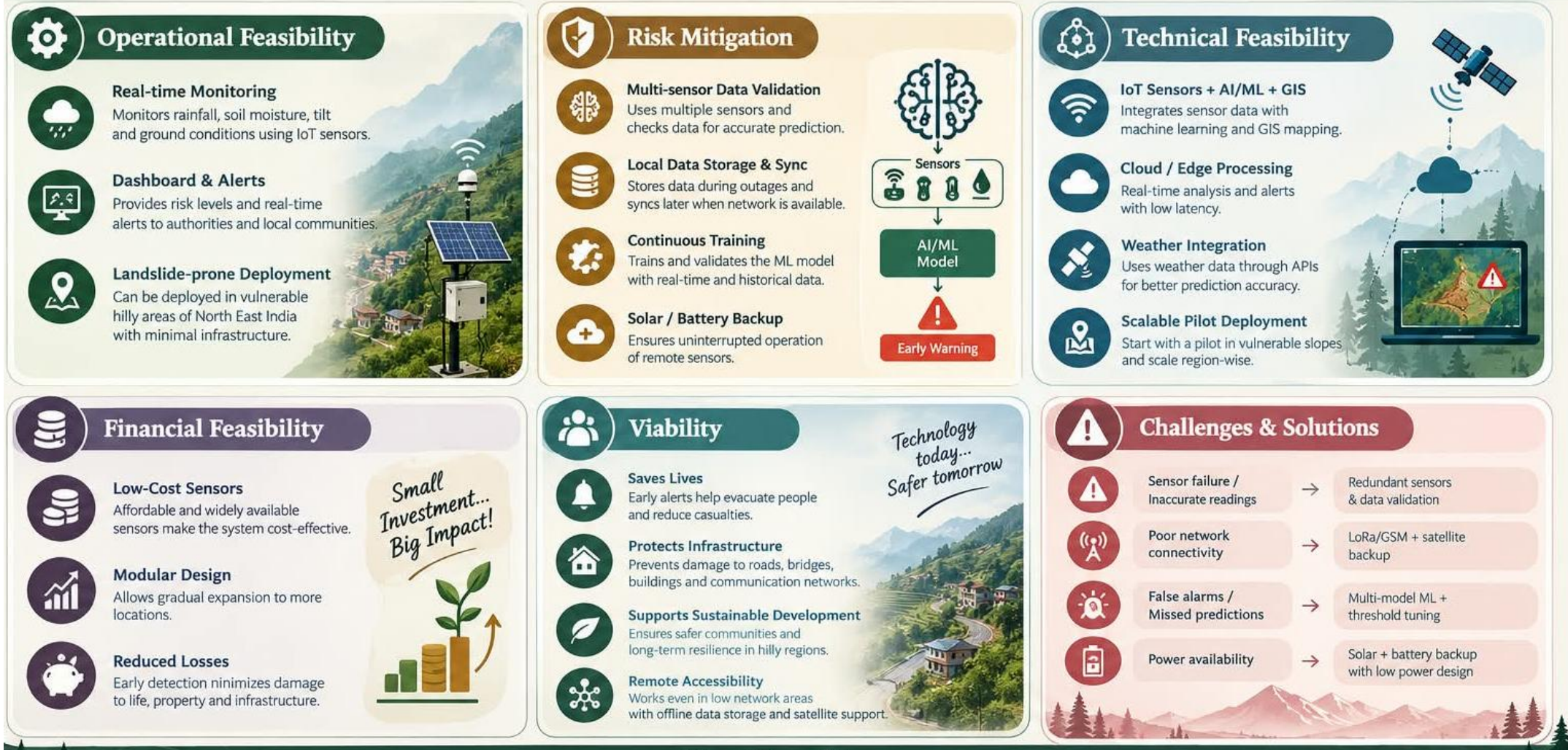
TerraSentinels | AI + IoT-Based Landslide Early
Warning System for Northeast India.

How It Works ?



System Flow / Technical Approach





IMPACT



BENEFITS

- Real-time Alerts**
Instant notifications via mobile app, SMS and local authorities.
- Accurate Risk Prediction**
AI analyzes sensor data (rainfall, soil moisture, ground movement) to predict landslide risk.
- Accessible to Remote Areas**
Works even in low network areas with offline data storage and satellite support.
- Community Empowerment**
Local training and awareness help people respond quickly and safely.
- Long-term Sustainability**
Builds climate resilience for a safer and stronger North East.

RESEARCH AND REFERENCES

- Details / Links of the reference and research work

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.